

# Open-loop $u(t)$ Experiment Report

Paper-setting Transformer control trained by PMP/KKT optimality gaps

This report focuses on the time-dependent open-loop control policy  $u_{\theta}(t)$ :  $[0,T] \rightarrow [0,u_{\max}]$ . The feedback case  $u(t,N)$  is intentionally excluded because it has different optimality conditions.

## 1. Model and Training Objective

We use the population dynamics and cost from the manuscript:

$$\frac{dN_i}{dt} = (r_i - \phi_i u(t) - M_i G(N(t))) N_i(t)$$

$$G(N) = \log(1 + (1/m) \sum_k N_k)$$

$$J(u) = \alpha^T N(T) + \int_0^T [\beta^T N(t) + \gamma u(t)] dt$$

Reported parameters:  $T=10$ ,  $m=21$ ,  $u_{\max}=3$ ,  $\alpha=1$ ,  $\beta=0.1$ ,  $\gamma=20$ , and  $N_i(0)=10$ . The control is represented by a small Transformer encoder over normalized time:  $u_{\theta}(t_k)=u_{\max} \sigma(g_{\theta}(t_k))$ .

Given  $u_{\theta}(t)$ , we roll out  $N_{\theta}(t)$ , solve the costate equation backward, and compute the switching function

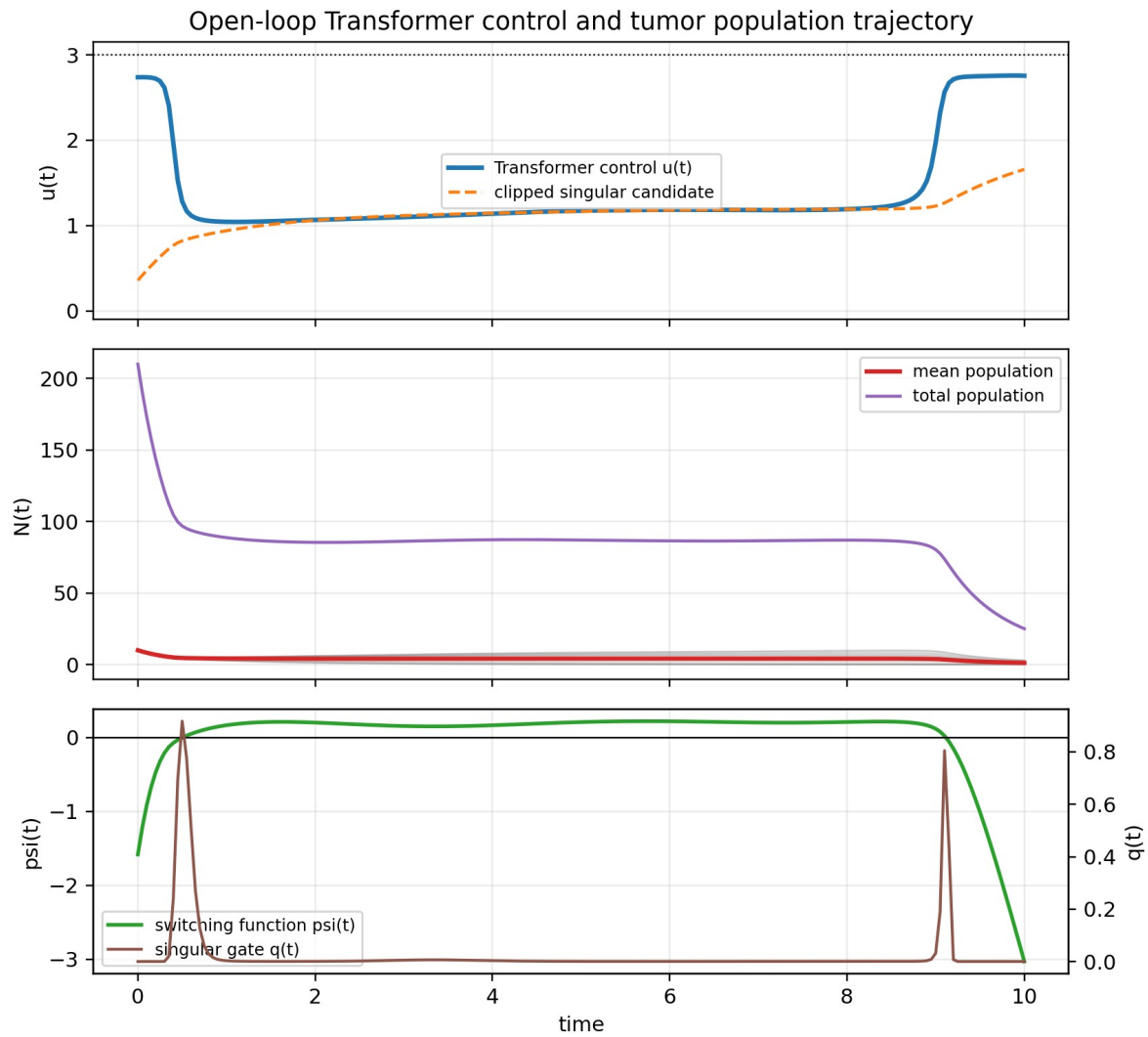
$$\psi(t) = H_u(N, \lambda, u) = \gamma - \sum_i \phi_i \lambda_i(t) N_i(t).$$

| component             | role                                                                                      |
|-----------------------|-------------------------------------------------------------------------------------------|
| non-singular KKT loss | enforces $u=0$ when $\psi>0$ and $u=u_{\max}$ when $\psi<0$                               |
| singular loss         | near $\psi=0$ , matches $u_{\theta}(t)$ to the singular candidate $u_{\text{sing}}(N(t))$ |

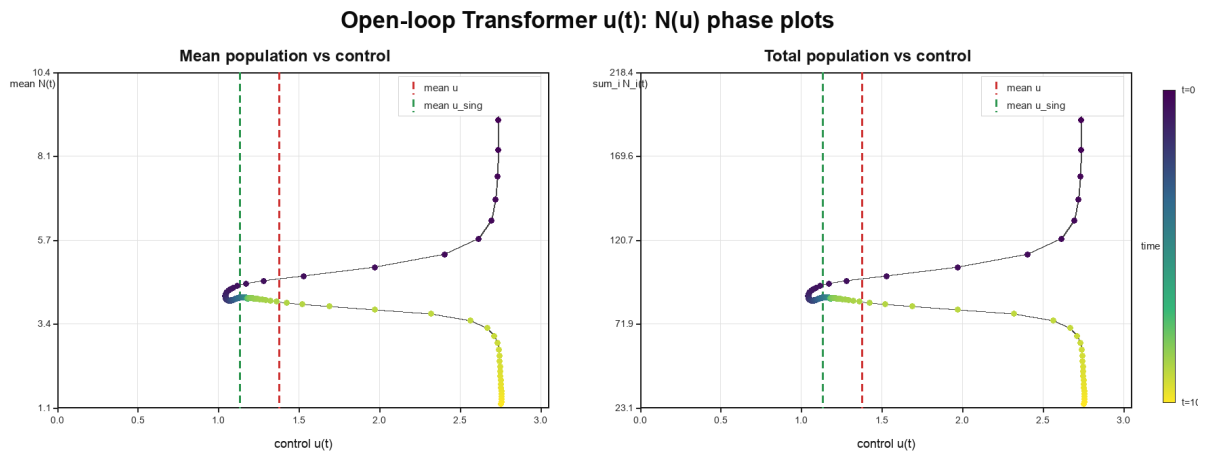
Here  $q(t)$  is a smooth weight that switches between the singular condition near  $\psi(t)=0$  and the boundary KKT condition away from  $\psi(t)=0$ .

## 2. Learned $u(t)$ , $N(t)$ , and $N(u)$

The learned control starts high, transitions to a lower interior/singular-like region, and increases again near the terminal portion.



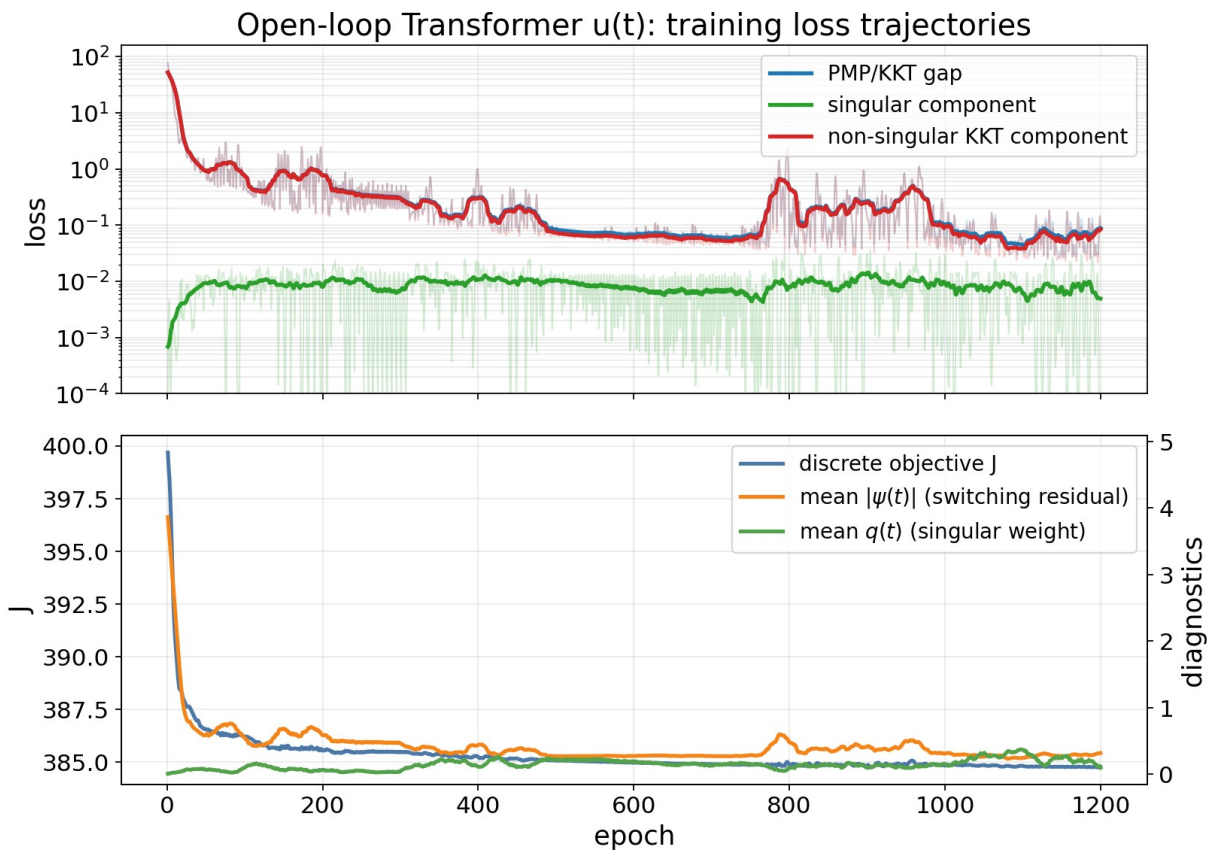
The  $N(u)$  phase plot shows population directly against the applied control value; color indicates time.



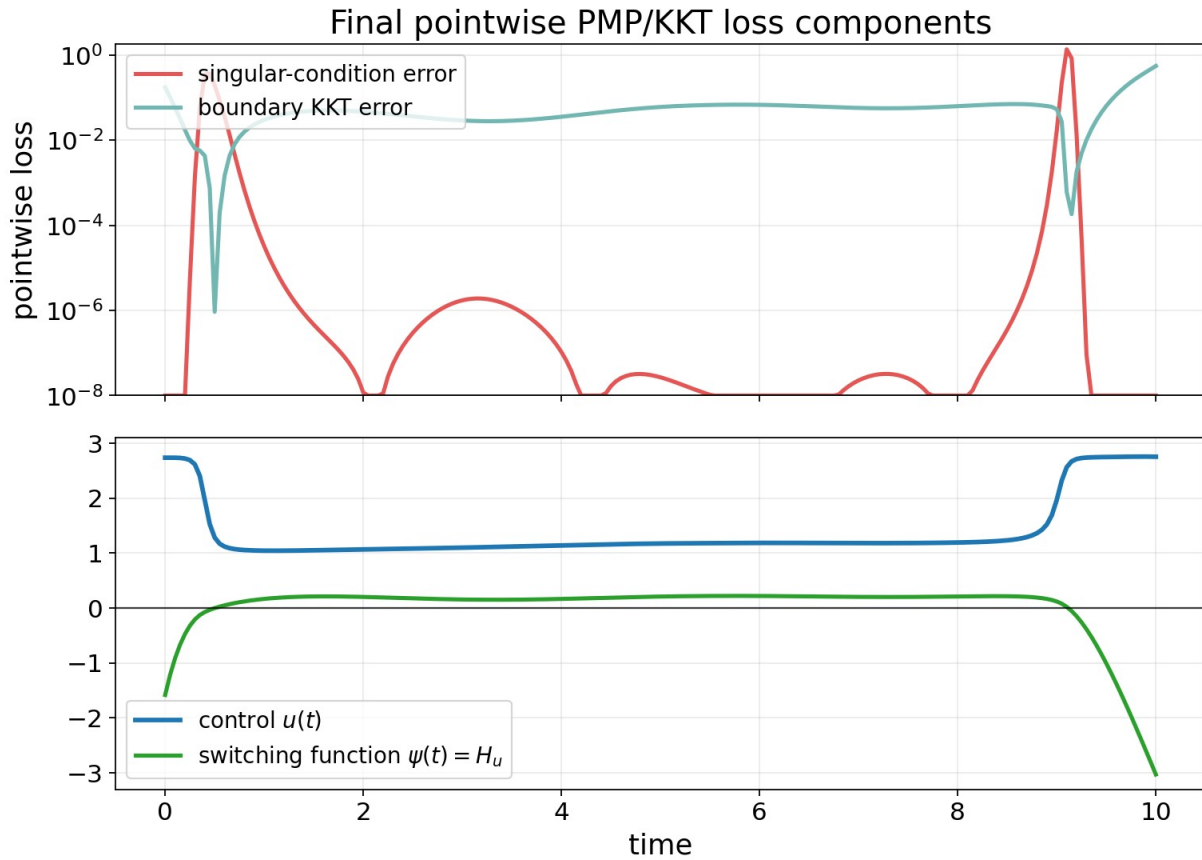
## 3. Training Loss Trajectories for PMP/KKT Conditions

The table reports the smallest recorded training loss. In this experiment, the training loss is the manuscript's PMP/KKT optimality gap, composed of the singular condition and the non-singular Hamiltonian minimization condition.

| metric                                              | value                            |
|-----------------------------------------------------|----------------------------------|
| total training loss / PMP-KKT optimality gap        | 0.02637                          |
| singular-condition training loss                    | 0.00167                          |
| non-singular Hamiltonian-minimization training loss | 0.02471                          |
| objective J on the training grid                    | 384.76                           |
| range and mean of $u_{\text{theta}}(t)$             | min 1.038, max 2.758, mean 1.372 |
| terminal mean state                                 | 1.207                            |



The final pointwise diagnostic below uses a smooth weight  $q(t)$  to separate the two regimes: near  $\psi(t)=0$  it emphasizes the singular-condition error; away from  $\psi(t)=0$  it emphasizes the boundary KKT error.



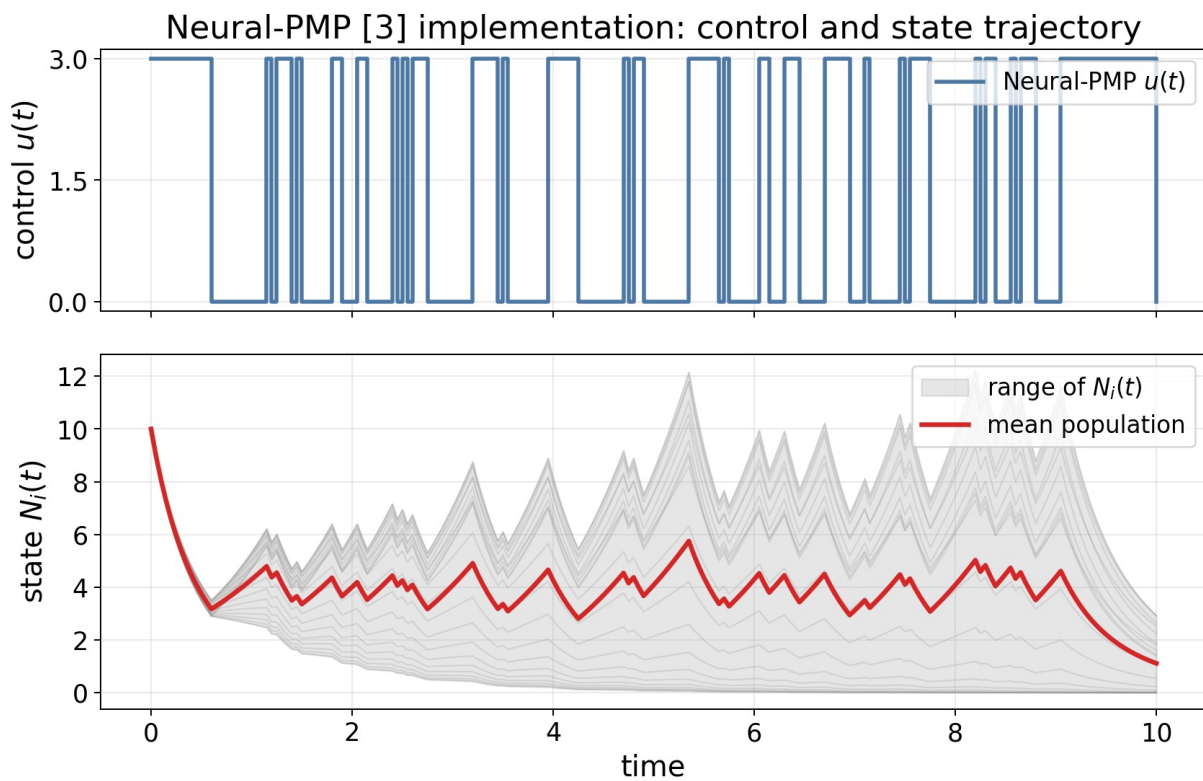
## 4. Comparison With Related Work [3]

Related work [3], Neural-PMP / PMP-gradient, also follows a Pontryagin-style procedure: forward state integration, backward costate recursion, and Hamiltonian-gradient updates of a discrete control sequence. For comparison on the same model and parameters, we implemented the corresponding control-update step from [3].

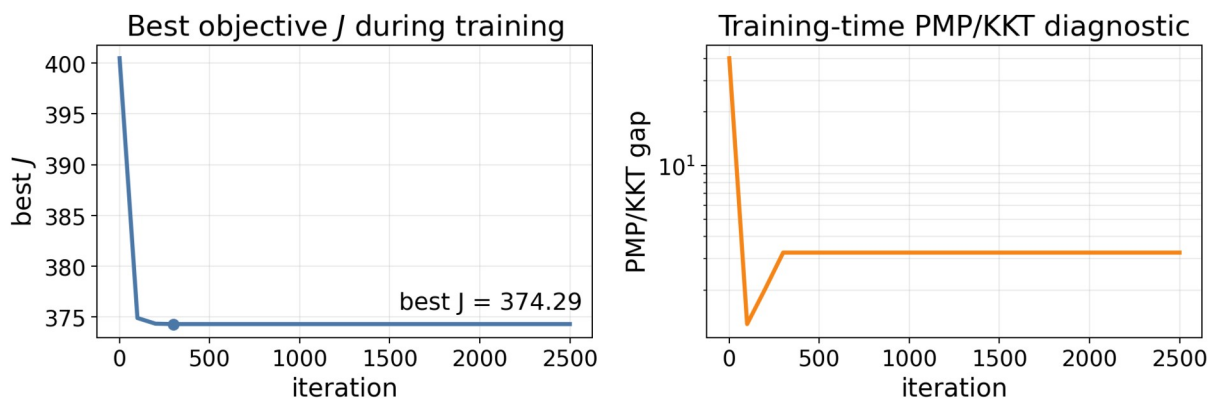
| method             | control update / training criterion             | PMP/KKT gap | objective $J^*$ | note                     |
|--------------------|-------------------------------------------------|-------------|-----------------|--------------------------|
| Transformer $u(t)$ | paper PMP/KKT optimality-gap loss               | 0.0523      | 386.70          | main $u(t)$ reproduction |
| Neural-PMP [3]     | Hamiltonian-gradient update of control sequence | 7.47        | 387.02          | related-work comparison  |
| direct grid cost   | minimize discretized $J$                        | 0.805       | 386.47          | reference only           |
| constant $u=1.5$   | no training                                     | 76.89       | 400.40          | scale check              |

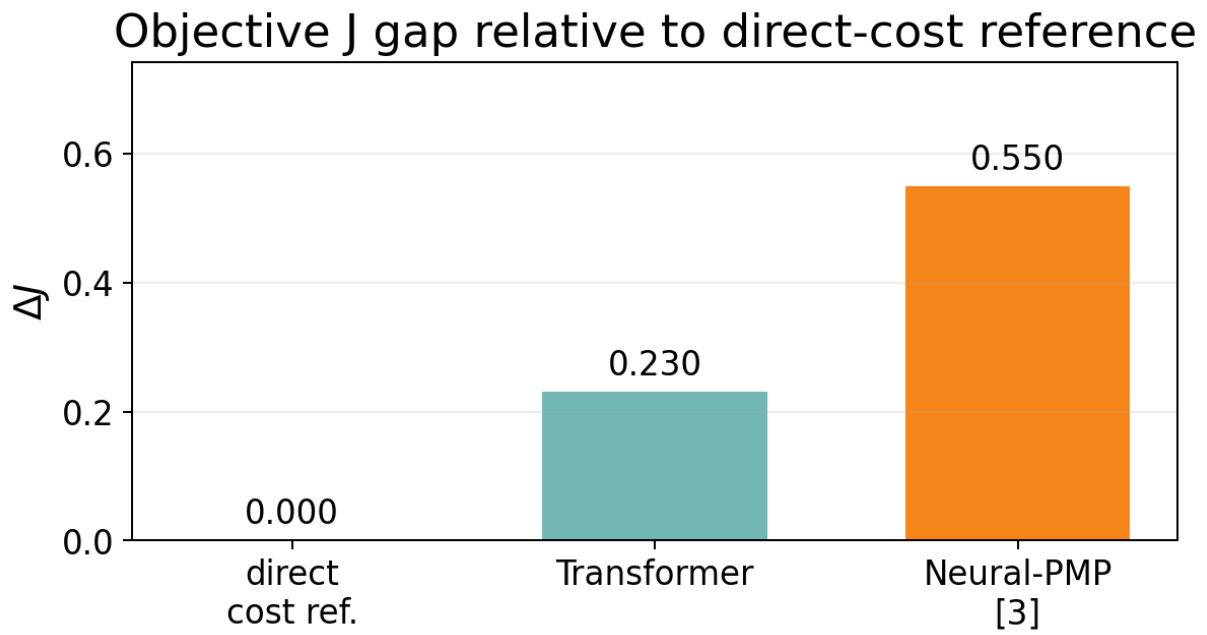
\* The  $J$  values in this table are computed after fixing  $u(t)$ , reintegrating  $N(t)$  with a finer-step fourth-order Runge-Kutta method, and then applying the manuscript objective definition. This is only to use the same numerical integration accuracy across methods.

Under the same PMP/KKT gap calculation, the Transformer  $u_{\theta}(t)$  has a much smaller gap than the Neural-PMP implementation of [3]. When  $J$  is recomputed with the same numerical evaluator, the Transformer also has a slightly lower objective value in this run.



Neural-PMP [3] implementation: selected-run training trajectories





## 5. Conclusion

The requested  $u(t)$  reproduction is complete. The Transformer control trajectory trained with the manuscript's PMP/KKT optimality-gap loss is smooth and satisfies the control bounds, reduces the training optimality gap from 76.26 to 0.02637, and gives  $J$  approximately 386.70 under the common numerical evaluation. The feedback case  $u(t, N)$  is not included here.